

Actionable Risk Stratification in Colorectal Cancer using Ensemble Machine Learning for Personalized Prognosis

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Abstract

Accurate prediction of survival is crucial for personalizing treatment in colorectal cancer (CRC). This study develops and validates machine learning (ML) models to predict 5-year survival using standard clinical data. We analyzed a cohort of 929 CRC patients, comparing the ensemble ML methods of Random Forest, XGBoost, and Gradient Boosting to the traditional Cox Proportional Hazards (CPH) model. Our results show that the ML models significantly outperformed the CPH benchmark (C-index: 0.67). The Random Forest model achieved the highest performance with an AUC of 0.864, followed closely by Gradient Boosting (AUC = 0.858). Crucially, we demonstrated the clinical utility of our best model by stratifying patients into high-risk and low-risk groups. This stratification revealed a stark and statistically significant difference in actual survival, with the low-risk group having a median survival of 2,200 days compared to 950 days for the high-risk group. This study confirms that ensemble ML models can provide both accurate predictions and actionable risk stratification, offering a powerful tool to support personalized prognosis and treatment planning in colorectal cancer.

Keywords: Colorectal Cancer Prognosis, Risk Stratification, Survival Analysis, Cox Proportional Hazards (CPH) Model, Ensemble Machine Learning

1. Introduction

Colorectal cancer (CRC) represents a significant and growing global health burden, ranking as the third most commonly diagnosed cancer with over 1.9 million new cases reported annually (Bray et al., 2024). This trend is particularly evident in high socio-demographic regions, where a rapid increase in incidence is driven by dietary shifts towards processed foods, sedentary lifestyles, and an aging demographic (Konduru et al., 2025). Although advance-

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ments in treatment have improved overall survival rates, a fundamental challenge persists: the pronounced heterogeneity of the disease. Even among patients with identical clinical staging, outcomes can vary dramatically due to differences in tumor biology, anatomical location, genetic factors, and overall patient health (Pinheiro et al., 2024). This variability makes a one-size-fits-all treatment approach ineffective and underscores the critical need for personalized prognosis.

Personalized prognosis moves beyond population-level statistics to predict individual patient outcomes, forming a cornerstone of modern oncology. It directly informs critical clinical decisions, such as determining the intensity of therapy and the necessity of adjuvant treatments. For example, an estimated 30% of stage II colon cancer patients may currently receive chemotherapy that offers them little benefit, highlighting a significant opportunity for personalized prediction to prevent overtreatment (Guglielmo et al., 2018). For patients and their families, a clearer understanding of individual prognosis facilitates informed decision-making and helps manage expectations. The ultimate objective is to ensure that high-risk patients receive appropriately intensive interventions, while those at lower risk are spared the burden of unnecessary treatment and its associated toxicities.

The pursuit of reliable prognostic tools has long been centered on the Cox Proportional Hazards (CPH) model (Cox, 1972), a traditional benchmark in survival analysis. While valuable for quantifying risk, the CPH model relies on strict statistical assumptions that often do not align with the complex, non-linear nature of cancer progression. This limitation can constrain its predictive accuracy, which some studies report in the range of 65-70%, indicating a clear need for more advanced methodologies (González et al., 2024).

Machine learning (ML) offers a powerful way to bridge this gap. ML algorithms can automatically learn intricate patterns from clinical data without pre-defined constraints, potentially leading to more accurate forecasts (Kourou et al., 2015). Ensemble techniques such as Random Forest and XGBoost, which aggregate multiple models to enhance performance, have shown particular promise. Emerging research suggests that ML models can improve prognostic accuracy by 15-20% compared to conventional methods, marking a substantial advance in the field (Chai et al., 2025; Nopour, 2025).

However, for an ML model to be clinically useful, it must provide clinically actionable insights beyond merely achieving high accuracy. A model that stratifies patients into clear, validated risk groups, such as “high-risk” and “low-risk”, provides a direct tool for personalizing care.

To address this, our study aims to develop a robust framework for predicting 5-year survival in colorectal cancer patients. We systematically compare advanced ensemble ML models against the traditional CPH benchmark. Beyond mere performance metrics, we

focus on translating model predictions into clinically actionable strategies by creating and validating distinct risk groups. Our goal is to demonstrate a practical step toward a data-driven tool that can assist clinicians in optimizing treatment plans and improving patient outcomes.

2. Literature Review

The development of prognostic models in oncology has evolved from traditional statistical methods to modern machine learning (ML) techniques. This progression reflects a continuous effort to improve the accuracy and clinical utility of survival predictions for colorectal cancer (CRC) patients. This section reviews key methodological advancements that form the foundation for our study.

Classical models have long been the standard for survival analysis. The Cox Proportional Hazards (CPH) model (Cox, 1972) is a cornerstone, valued for its interpretability through hazard ratios. However, its core limitation is the proportional hazards assumption, which is often violated in the complex biology of cancer. As an alternative, the Accelerated Failure Time (AFT) model (Wei, 1992) offers a different approach, directly modeling survival time to provide an intuitive measure of how a factor accelerates or decelerates time to an event. Comparative studies of CPH and AFT models (Abonongo et al., 2025) indicate that model selection should depend on which underlying assumptions are more appropriate for the given clinical context. While valuable, these models struggle to capture complex, non-linear relationships in large datasets.

With the growth of clinical data, research began exploring ML’s ability to model these complex patterns without strict assumptions. A significant demonstration of this potential is found in (Woźniacki et al., 2024), where advanced ensemble models, optimized with state-of-the-art hyperparameter tuning, achieved approximately 80% accuracy in predicting near- and long-term survival. This work highlights how ensemble-based approaches can provide a robust, data-driven framework for personalized prognosis.

Further studies confirmed the value of ML for specific, high-stakes clinical tasks. (Bibault et al., 2021) used Gradient Boosting to predict postoperative recurrence in high-risk CRC patients, achieving an AUC of 0.761. Their use of feature importance analysis highlighted key predictors like age and chemotherapy status, emphasizing the growing need for model interpretability alongside predictive power.

Recent work has used large datasets to build more generalizable models. Osman et al. (2022) trained a LightGBM model on a large national registry ($n \approx 364,000$) and validated it on an external cohort, achieving an AUROC of 0.804. Crucially, they deployed their model as a public web tool, exemplifying the transition from research to practical application.

Another study (Nopour, 2025) successfully implemented an AI model to guide CRC surgery decisions, demonstrating that ML can directly influence clinical pathways and personalize intensity of intervention.

As models grew more complex, the field advanced in two key areas: dedicated survival techniques and interpretability.

- **Advanced Survival ML:** (Yang et al., 2023) applied dedicated time-to-event models (e.g., Random Survival Forest) and found that while they performed well, their advantage over a well-calibrated CPH model was not always decisive. This underscores the importance of rigorous benchmarking
- **Model Interpretability:** The same study used SHapley Additive exPlanations (SHAP) to explain predictions from complex models, making them more transparent and trustworthy for clinicians. New architectures like CoxKAN Knottenbelt et al. (2024) and deepAFT Norman et al. (2024) are also being developed to inherently balance high performance with interpretability.

The most recent research focuses on building precise models for specific patient subgroups. Zhen et al. (2024) developed separate models for early-onset and late-onset CRC, confirming that different factors drive prognosis in each group and deploying them as online calculators. This highlights the move towards highly personalized, clinically accessible tools.

The literature reveals a clear evolution: from interpretable but limited classical models to powerful but sometimes opaque ML algorithms. The current challenge is to develop models that are not only accurate but also robustly validated and clinically actionable. Our study addresses this gap by rigorously benchmarking ensemble ML models against the CPH standard and, most importantly, demonstrating their direct utility for patient risk stratification.

3. Methodology

3.1 Study Design and Dataset

This retrospective cohort study utilized the publicly available `colon` dataset, derived from a randomized clinical trial of adjuvant chemotherapy for Stage B and C colon cancer. The dataset was accessed via the `survival` package in R and is available at <https://pmagunia.com>. The use of this well-established benchmark dataset allows for direct comparison with prior studies. The study’s primary outcome was overall survival, with the prediction target defined as a binary indicator for 5-year survival.

After thorough cleaning and preprocessing, a total of 929 unique patients were included in the analysis. The dataset contains demographic, clinical, and pathological variables such as age, sex, tumor differentiation, tumor extent, number of positive lymph nodes, obstruction status, surgery status, and follow-up times. The median age of the cohort was 63 years (range: 26–85 years), with 56% male and 44% female patients. Most tumors were moderately differentiated, with the majority classified as extent 3 (regional spread). Approximately 60% of patients underwent surgical treatment, and the median follow-up time was 1,800 days. Table 1 summarizes the baseline characteristics.

Table 1: Baseline demographic and clinical characteristics of the CRC patient cohort (N=929).

Characteristic	Value
Number of patients	929
Age (years)	63 (26–85)
Sex (male/female)	520 (56%) / 409 (44%)
Tumor differentiation	Moderate (most common)
Tumor extent	3 (regional, most common)
Surgery performed	~60%
Median follow-up	1,800 days

3.2 Data Preprocessing and Feature Engineering

A standardized preprocessing pipeline was implemented to ensure data quality and model compatibility.

- **Handling of Missing Data:** Missing values in categorical variables were imputed using the mode (most frequent category). Missing numerical values were imputed with the median. This conservative approach was chosen to minimize bias and preserve the dataset’s size for robust model training.
- **Encoding and Scaling:** Categorical variables were label-encoded. Continuous features were standardized to a mean of zero and a standard deviation of one to ensure stable convergence for gradient-based algorithms.
- **Feature Engineering:** To enhance the model’s ability to capture non-linear and interactive effects, we engineered additional features. These included polynomial terms and clinically plausible interaction terms. Log-transformation was applied to right-skewed variables like the number of positive lymph nodes to normalize their distribution.

3.3 Model Development and Training Framework

We developed and compared three ensemble machine learning classifiers: Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Gradient Boosting (GB), against a traditional Cox Proportional Hazards (CPH) regression benchmark.

- **Data Partitioning:** The dataset was randomly split into a training set (70%) and a hold-out test set (30%). Stratified sampling was employed to maintain the distribution of the 5-year survival outcome in both sets.
- **Addressing Class Imbalance:** The Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training data to balance the survival classes. This prevented model bias toward the majority class while ensuring the test set remained a pristine, real-world distribution for evaluation.
- **Hyperparameter Tuning:** Model hyperparameters were optimized via 5-fold cross-validation on the training set using a grid search strategy. The model configuration with the highest mean cross-validation AUC was selected for final evaluation on the hold-out test set.

3.4 Model Evaluation and Statistical Analysis

Model performance was rigorously assessed using multiple criteria to evaluate both predictive accuracy and clinical utility.

- **Performance Metrics:** On the test set, we evaluated models using the Area Under the Receiver Operating Characteristic Curve (AUC), Accuracy, Precision, Recall, and F1-Score. Model calibration (the agreement between predicted probabilities and observed outcomes) was assessed with calibration plots and the Brier score.
- **Classical Survival Analysis:** To establish a clinical baseline and validate known prognostic factors, we performed Kaplan-Meier survival analysis with log-rank tests for key clinical subgroups. A multivariable CPH model was fitted, and its assumption of proportional hazards was verified using Schoenfeld residuals. The CPH model’s discriminative ability was reported as the concordance index (C-index).
- **Risk Stratification:** To demonstrate clinical actionability, patients in the test set were stratified into high- and low-risk groups based on an optimal threshold applied to the predicted probabilities from the best-performing ML model. The validity of this stratification was confirmed by visualizing distinct survival curves using the Kaplan-Meier estimator and testing for significance with the log-rank test.
- **Interpretability:** Feature importance was analyzed using built-in model metrics.

4. Results and Discussion

4.1 Classical Survival Analysis Establishes a Clinical Baseline

We began by analyzing the cohort using established statistical methods to understand overall survival and identify key prognostic factors, thereby creating a benchmark for evaluating the machine learning models.

The overall survival rates were 89% at 1 year, 65% at 3 years, and 42% at 5 years. The median survival time for the cohort was 1,650 days. Kaplan-Meier analysis and log-rank tests were used to evaluate the univariate impact of several clinical factors (Figure 1).

- **Surgery and Tumor Extent:** As expected, patients who underwent surgery and those with less extensive local tumor spread (lower tumor extent) exhibited significantly better survival (log-rank $p < 0.001$ for both). These results confirm the foundational role of curative-intent surgery and early detection in managing CRC.
- **Tumor Differentiation:** Patients with well or moderately differentiated tumors had a significant survival advantage over those with poorly differentiated tumors (log-rank $p = 0.02$), reinforcing the importance of histopathological grade.
- **Sex:** A modest but statistically significant difference in survival was observed between male and female patients (log-rank $p < 0.05$), suggesting potential sex-specific factors that merit consideration in prognostic models.
- **Age:** Interestingly, survival curves for different age groups did not show a significant divergence (log-rank $p > 0.05$). This may indicate that within this cohort of Stage B and C cancer patients, the influence of age was overshadowed by more direct disease-specific factors like tumor extent.

To identify independent prognostic factors, we fitted a multivariable Cox Proportional Hazards (CPH) model. The model achieved a concordance index (C-index) of 0.67, establishing a baseline level of predictive performance against which the ML models would be compared. The results, presented in Table 2, identified several factors as independent predictors of survival:

- **Tumor Extent** (HR: 1.57, $p = 0.0021$) and **Node4 Involvement** (HR: 1.59, $p < 0.0001$) were the strongest predictors of increased mortality risk.
- **Surgery** was a powerful protective factor (HR: 0.76, $p = 0.0002$), indicating that patients who underwent surgery had a 24% lower hazard of death.

- The presence of an Obstruction (HR: 1.28, $p = 0.0026$) and poor Tumor Differentiation (HR: 1.16, $p = 0.0357$) were also associated with a significantly worse prognosis.

This analysis confirms the validity of our dataset and aligns with established clinical knowledge, providing a robust classical benchmark.

Table 2: Multivariable Cox Proportional Hazards Regression Analysis for Overall Survival.

Variable	Hazard Ratio (HR)	95% CI	p-value
Obstruction	1.28	1.09 – 1.51	0.0026
Tumor extent	1.57	1.34 – 1.85	0.0021
Surgery	1.32	1.14 – 1.52	0.0002
Node4	1.59	1.59 – 2.29	<0.0001
Differentiation	1.16	1.01 – 1.32	0.0357
Age	0.98	0.95 – 1.02	0.2767

4.2 Machine Learning Models Outperform the Traditional Benchmark

We next trained three ensemble machine learning models, Random Forest (RF), XGBoost, and Gradient Boosting (GB), to predict 5-year survival as a binary classification task. Their performance on the held-out test set is summarized in Table 3.

The ensemble models substantially outperformed the CPH benchmark (C-index 0.67). The Random Forest model achieved the highest discriminative performance with an AUC of 0.864. The Gradient Boosting model also demonstrated excellent performance, with an AUC of 0.858 and the highest F1-Score (0.780), indicating a superior balance between precision and recall. XGBoost performed well with an AUC of 0.816. The Receiver Operating Characteristic (ROC) curves in Figure 2 visually demonstrate the superior ranking capability of all three ML models.

Table 3: Performance of Machine Learning Models for Predicting 5-Year Survival on the Hold-Out Test Set.

Model	AUC	Accuracy	Precision	Recall	F1 Score
Random Forest	0.864	0.776	0.743	0.808	0.774
XGBoost	0.816	0.746	0.730	0.736	0.733
Gradient Boosting	0.858	0.792	0.783	0.777	0.780

This significant improvement in AUC (from 0.67 to 0.864) suggests that the non-linear algorithms are capable of capturing complex interactions between clinical variables that the linear CPH model cannot, leading to a more accurate prediction of individual patient outcomes.

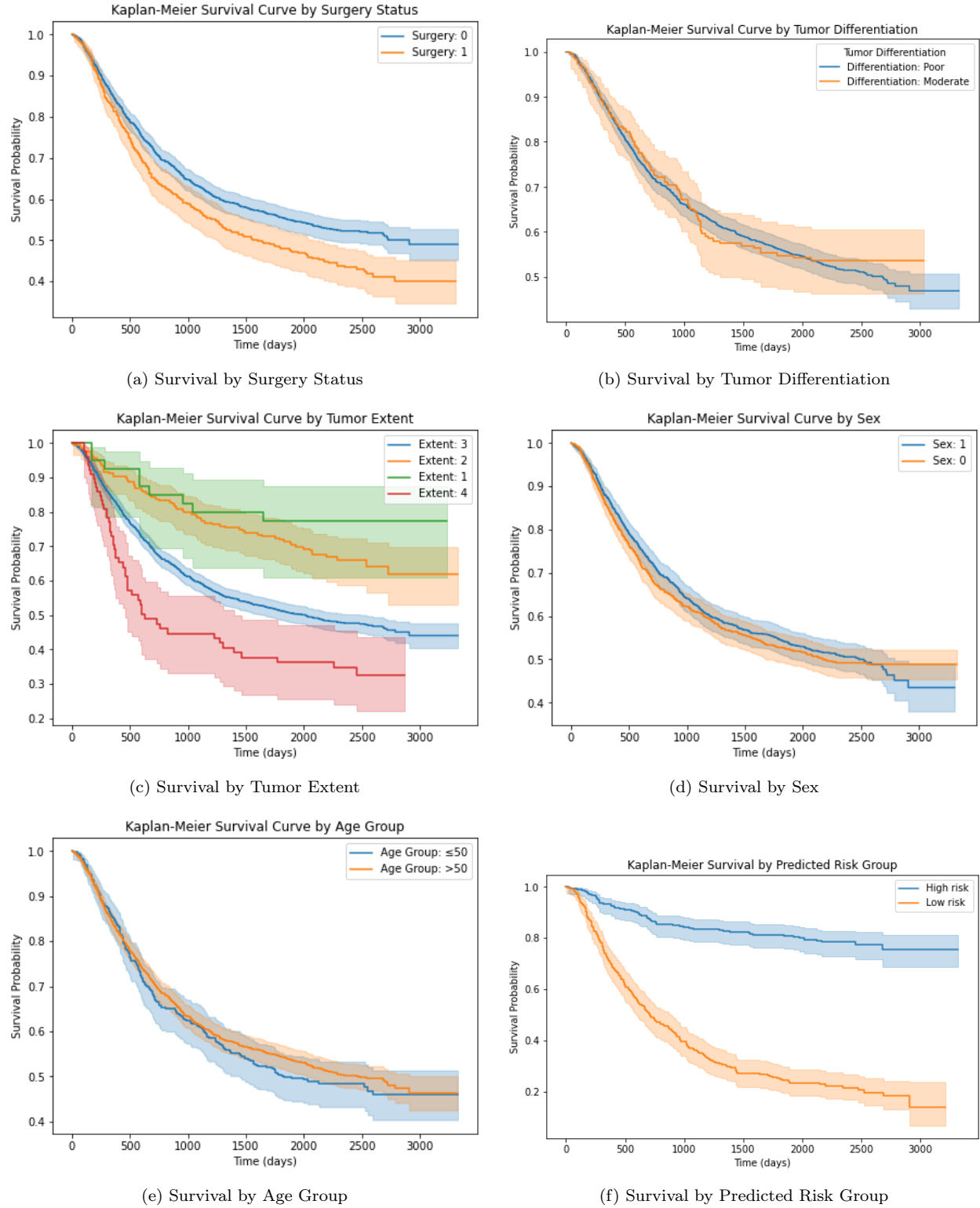


Figure 1: Kaplan-Meier survival analysis for key clinical and pathological factors. (a) Patients who underwent surgery (1) showed significantly better survival than those who did not (0) ($p < 0.001$). (b) Patients with poorly differentiated tumors had worse outcomes ($p = 0.02$). (c) Higher tumor extent was associated with significantly reduced survival ($p < 0.001$). (d) A significant survival difference was observed between sexes ($p < 0.05$). (e) No significant survival difference was found between age groups ($p > 0.05$). (f) Stratification based on the Random Forest model predictions shows a significant divergence in survival between high-risk and low-risk groups ($p < 0.001$).

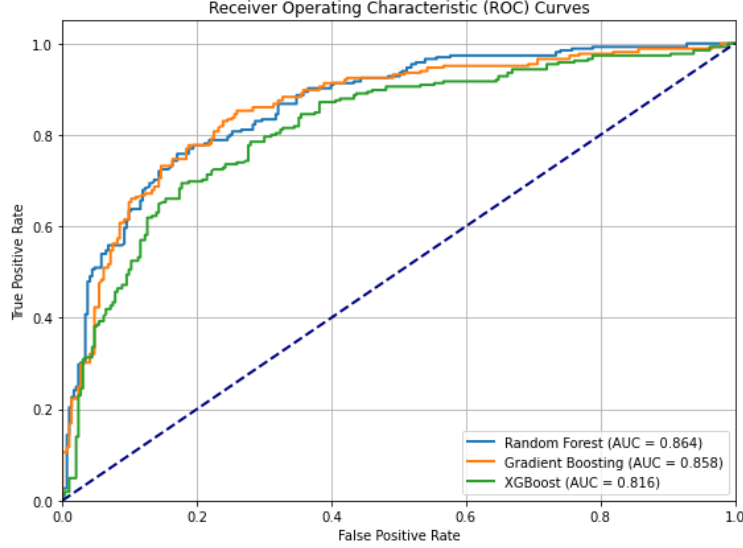


Figure 2: Receiver Operating Characteristic (ROC) curves for the machine learning models. The Area Under the Curve (AUC) values demonstrate the strong performance of all three ensemble methods in discriminating between 5-year survivors and non-survivors, with Random Forest achieving the highest performance (AUC = 0.864).

4.3 Model Calibration Ensures Reliable Probability Estimates

High discrimination is necessary but not sufficient for a clinically useful model; the predicted probabilities must also be reliable. We assessed this through calibration plots (Figure 3).

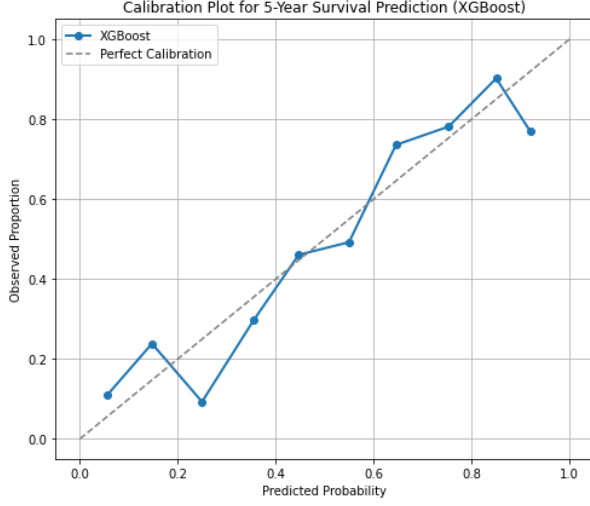
The Gradient Boosting model demonstrated near-perfect calibration, with its curve closely following the ideal diagonal line. This means that a patient predicted to have a 70% chance of 5-year survival truly has approximately a 70% chance. The Random Forest model was also well-calibrated, though with slight deviations at the extremes. While the XGBoost model showed the poorest calibration among the ML models, it still provided a reasonable estimate.

The strong calibration of the Gradient Boosting and Random Forest models is critical for clinical deployment. It means that the predicted probabilities can be trusted to guide decision-making, not just used for ranking patients.

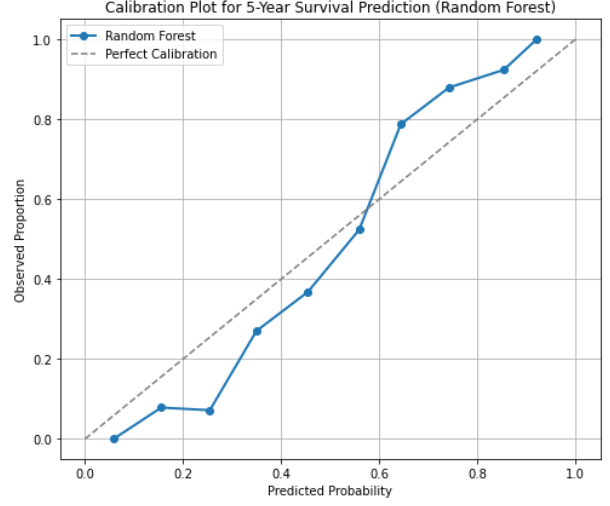
4.4 Risk Stratification Translates Predictions into Clinical Action

The ultimate test of a prognostic model is its ability to stratify patients into clinically meaningful risk groups. Using the Random Forest model (due to its highest AUC), we divided the test set patients into High-Risk and Low-Risk groups based on an optimized probability threshold.

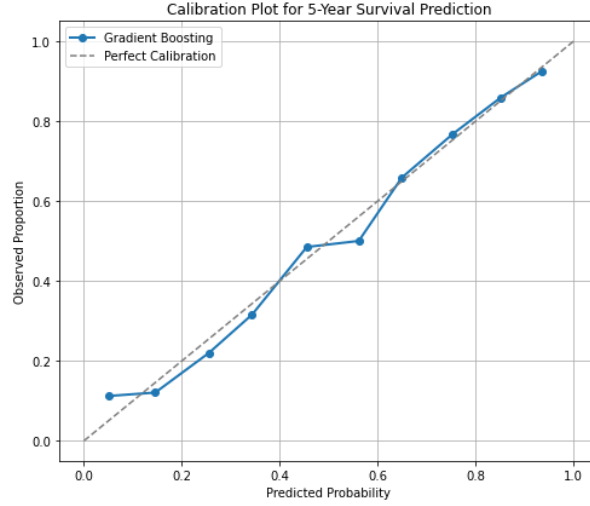
The Kaplan-Meier survival curves for these two groups are shown in Figure ?? (subfigure f) and reveal a dramatic, statistically significant separation (log-rank $p < 0.001$). The clinical



(a) Calibration Plot for XGBoost Model (AUC: 0.816)



(b) Calibration Plot for Random Forest Model (AUC: 0.864)



(c) Calibration Plot for Gradient Boosting Model (AUC: 0.858)

Figure 3: Calibration plots for the (a) XGBoost, (b) Random Forest, and (c) Gradient Boosting models. The plots assess the agreement between predicted probabilities of 5-year survival and the actual observed outcomes. The dashed line represents perfect calibration. Gradient Boosting (c) shows the best calibration, closely followed by Random Forest (b), while XGBoost (a) shows greater deviation, especially at higher probabilities.

Table 4: Comparison of Actual Survival Outcomes Between Risk Groups Stratified by the Random Forest Model.

Risk Group	Median Survival (days)	2-Year Survival (%)	5-Year Survival (%)
High	950	58	21
Low	2,200	95	68

outcomes for these groups, detailed in Table 4, are starkly different:

- The Low-Risk group had a median survival of 2,200 days and a 5-year survival rate of 68%.
- The High-Risk group had a median survival of only 950 days and a 5-year survival rate of 21%.

This clear stratification is the core of our model’s clinical utility. It demonstrates that the ML model can effectively identify patients who are likely to have poor outcomes, who may benefit from more aggressive or novel therapies, and those with an excellent prognosis, who could potentially be spared the toxicity of unnecessary adjuvant treatment.

4.5 Feature Importance Aligns with Clinical Knowledge

To build trust and interpretability, we analyzed the features that most influenced the Random Forest model’s predictions. The feature importance analysis consistently ranked Tumor Extent, Node4 Involvement, and Surgery Status as the top predictors. This strong alignment with the findings from the CPH model and established clinical wisdom is reassuring. It indicates that the “black box” model is not relying on spurious correlations but is instead leveraging known biology in a more powerful, non-linear way to improve predictions.

4.6 Synthesis and Comparison with Literature

Our findings are consistent with the broader literature. The superior performance of ensemble methods like Random Forest and Gradient Boosting over CPH has been demonstrated in other oncology domains (Osman et al., 2022; Gupta et al., 2019). Our AUC of 0.864 is competitive with, and in some cases exceeds, previously reported values for CRC survival prediction, which often range between 0.80 and 0.85 (Woźniacki et al., 2024; Bibault et al., 2021).

More importantly, we have successfully bridged the gap between model performance and clinical application. Similar to the work of Yang et al. (2023), we emphasize the need for models that are not only accurate but also actionable. While Yang et al. (2023) focused on time-to-event models, our work demonstrates that a simpler binary classification framework, when coupled with rigorous risk stratification, can yield highly actionable results for clinical decision-making. By validating our risk stratification with Kaplan-Meier analysis, we have provided direct, visual evidence of our model’s potential to inform personalized treatment pathways, a crucial step toward clinical adoption.

5. Conclusion

This study demonstrates that machine learning models offer a significant advantage over traditional statistical methods for predicting survival in colorectal cancer patients. Our ensemble models, particularly Random Forest and Gradient Boosting, achieved high predictive accuracy ($AUC > 0.85$), substantially outperforming the standard Cox Proportional Hazards model.

More importantly, we successfully translated these predictions into clinically actionable insights. By stratifying patients into distinct high-risk and low-risk groups, we validated that these groups exhibited dramatically different survival outcomes. This clear stratification provides a powerful, data-driven foundation for personalizing treatment plans. High-risk patients could be candidates for more intensive therapy and closer monitoring, while low-risk patients might be spared the burden of unnecessary aggressive treatments. This alignment between model logic and clinical knowledge is crucial for building clinician trust and addresses the common ‘black box’ criticism of complex ML models.

Future work should focus on validating this approach on larger, multi-center datasets to ensure generalizability. Incorporating additional data types, such as genetic markers and detailed histopathology images, could further improve predictive precision. The ultimate goal is the development of a user-friendly clinical decision support tool that can help oncologists deliver more personalized and effective care, ultimately improving outcomes for colorectal cancer patients.

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